

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

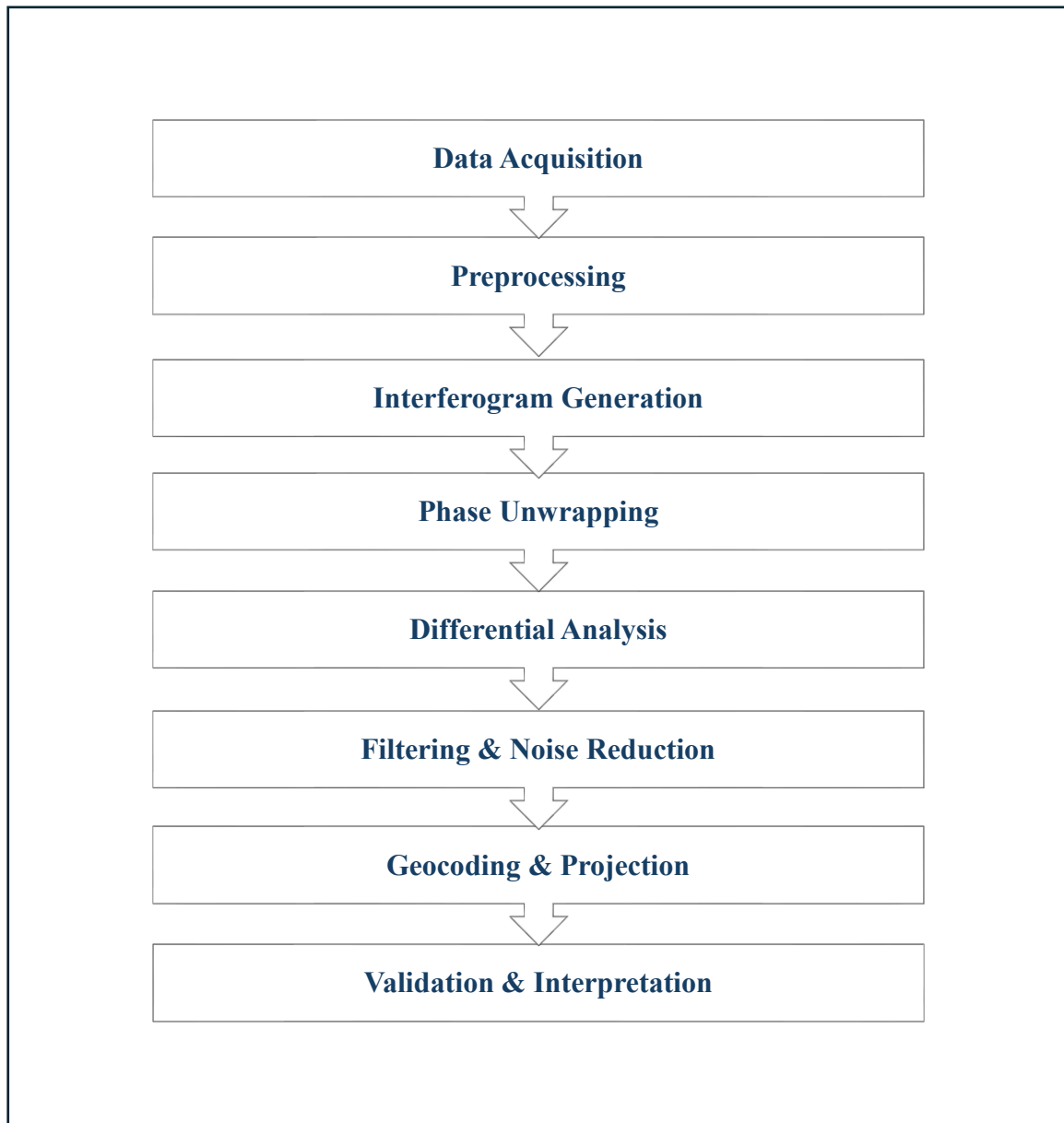
Multi-temporal Sentinel-1 SLC IW mode data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
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S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
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S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan
Bridge No. – 899/1
Bridge Chainage – 913+100

Project - Chittorgarh to Kota
Zone – RJ 06
Structure Type – VUP

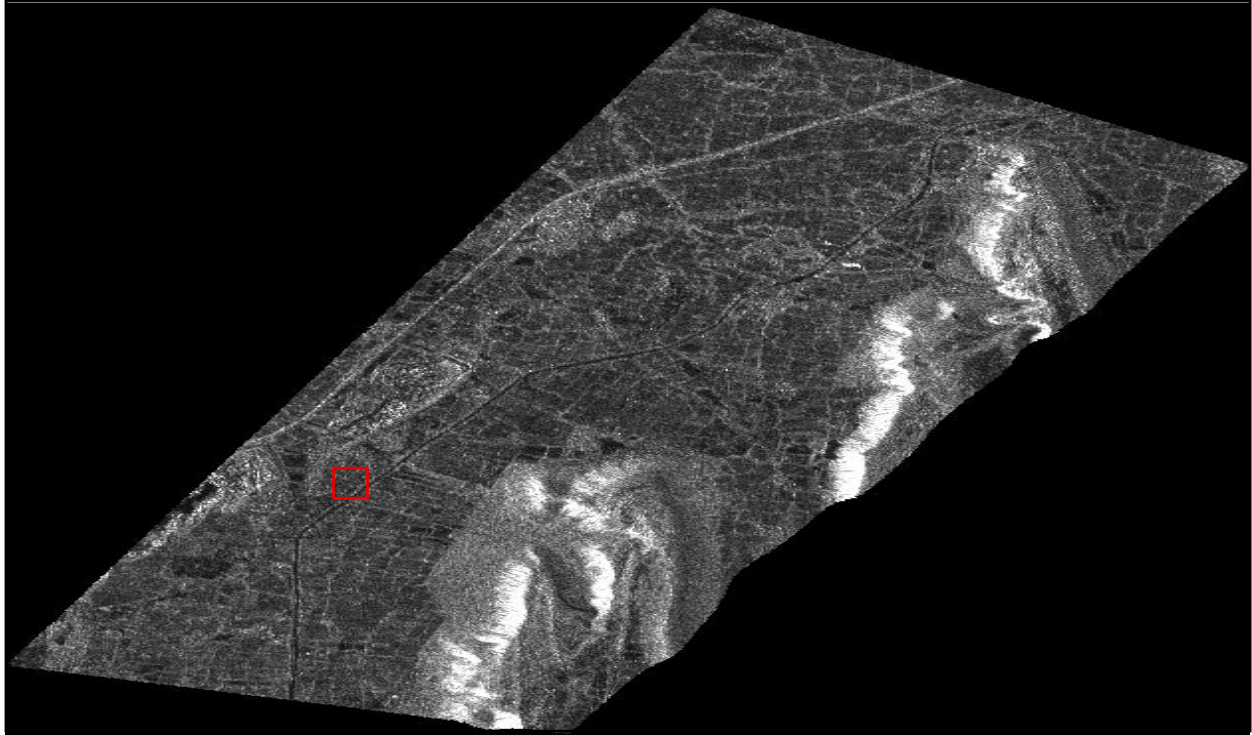
Sar Parameters

Sensor – Sentinel-1 (C-Band)
Beam Mode – IW (Interferometric Wide Swath)
Path – 34
Frame – 519

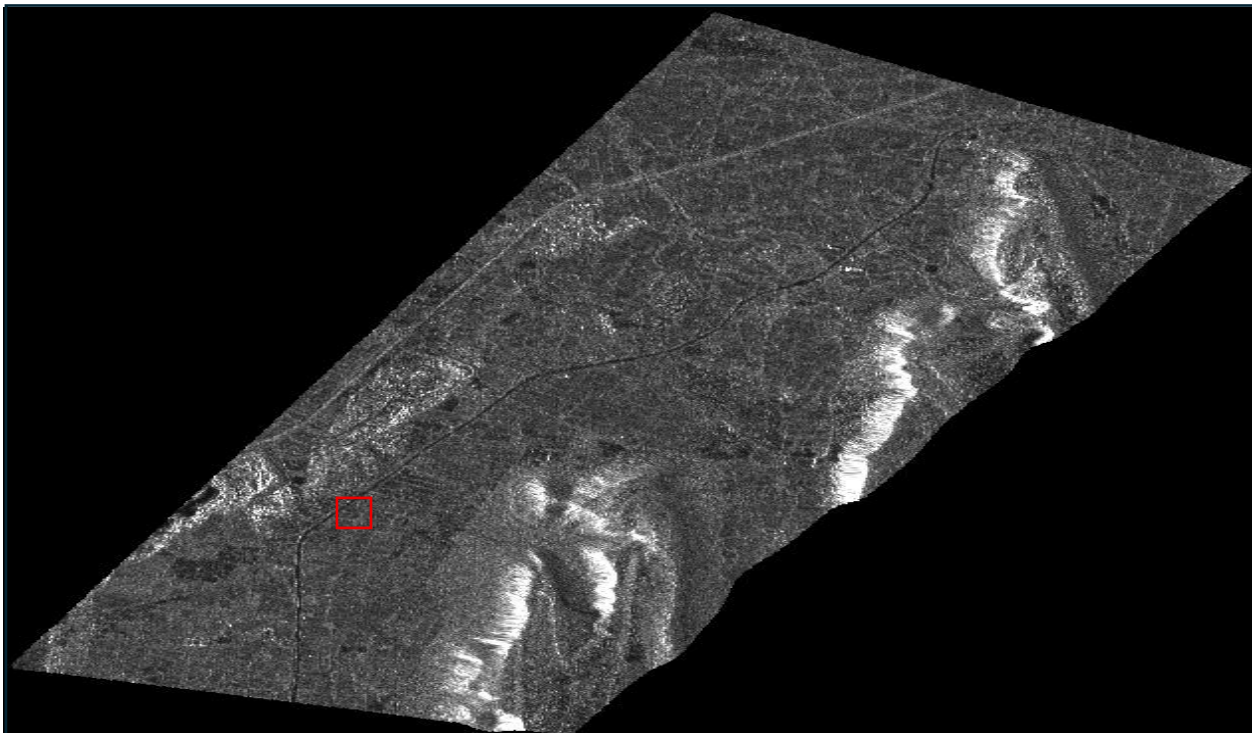
Flight Direction - DESCENDING
Polarization – VV+VH
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

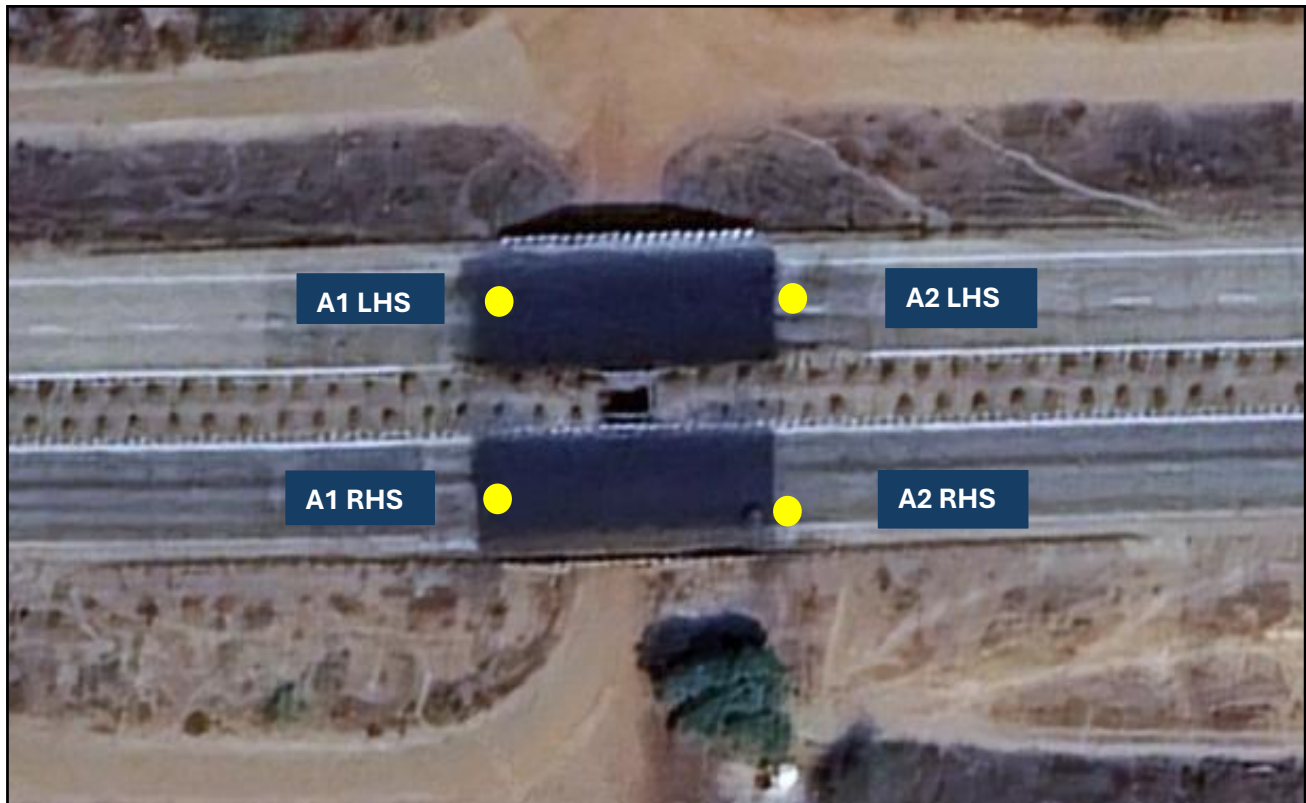


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

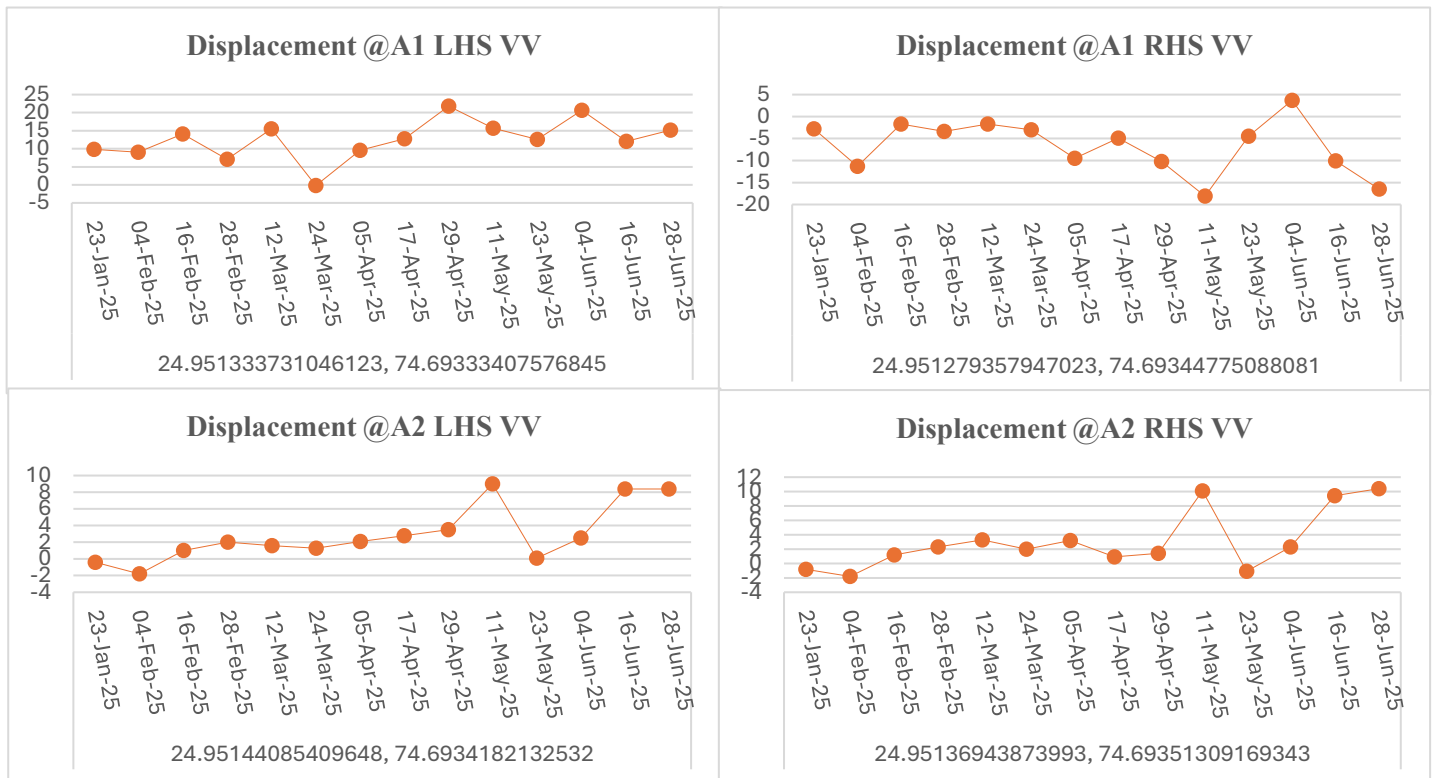
VUP – CH-913+100

Length – 22.49M

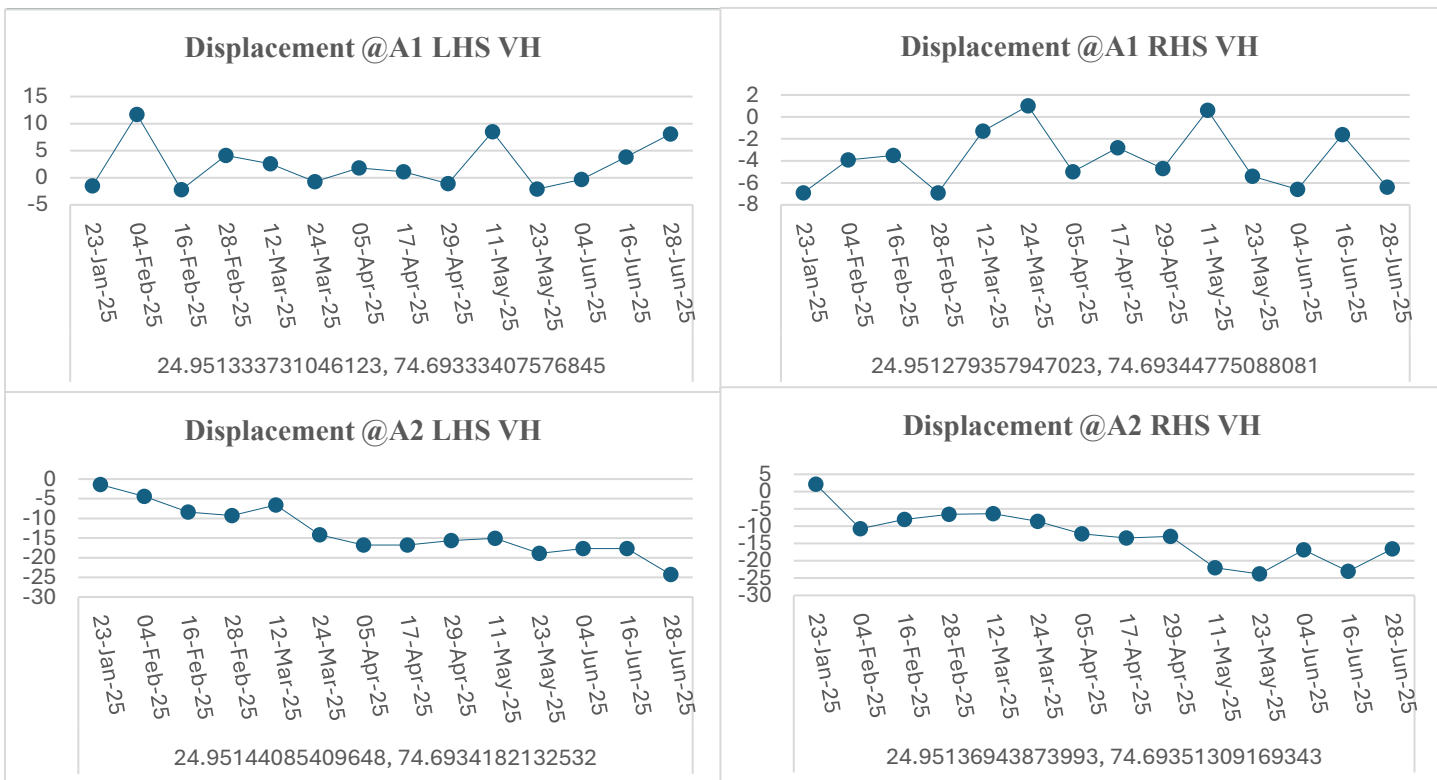


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1	Abutment 1		
A2	Abutment 2		
LHS	Left Hand Side		
RHS	Right Hand Side		

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



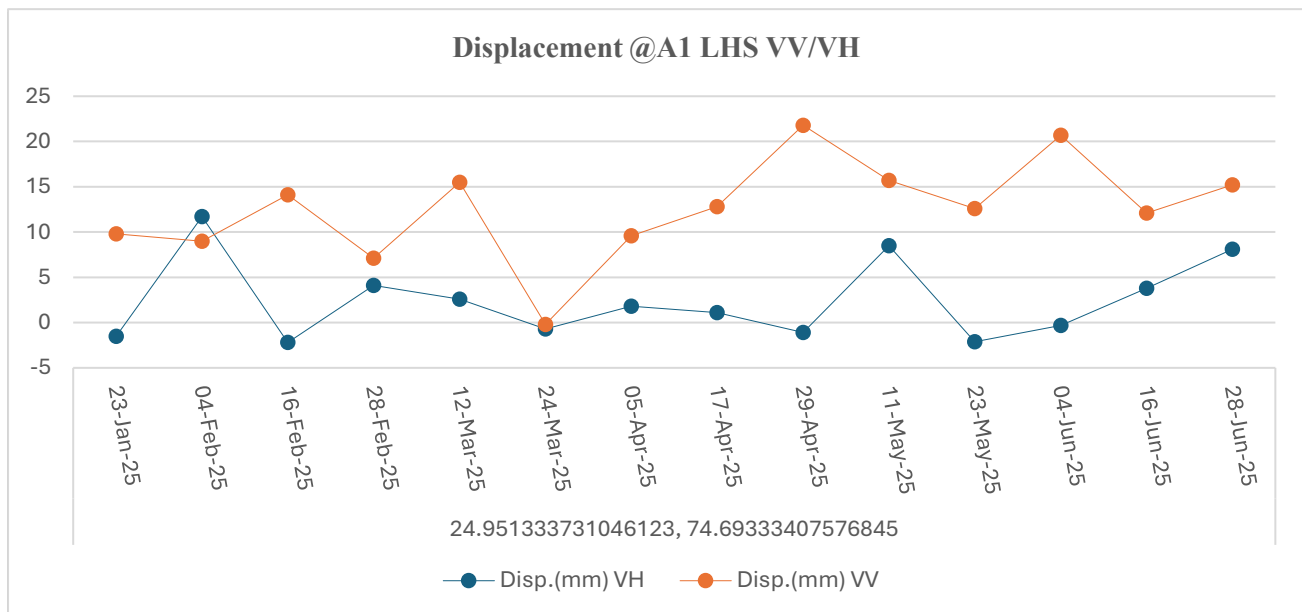
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (24.951333731046123, 74.69333407576845)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** VV exhibits a consistently positive trend, ranging roughly from +7 mm to +22 mm, indicating predominant upward movement with noticeable temporal variability.
- **Horizontal Displacement (VH)** values at A1 LHS range approximately from –3 mm to +12 mm, showing moderate oscillatory lateral movement.

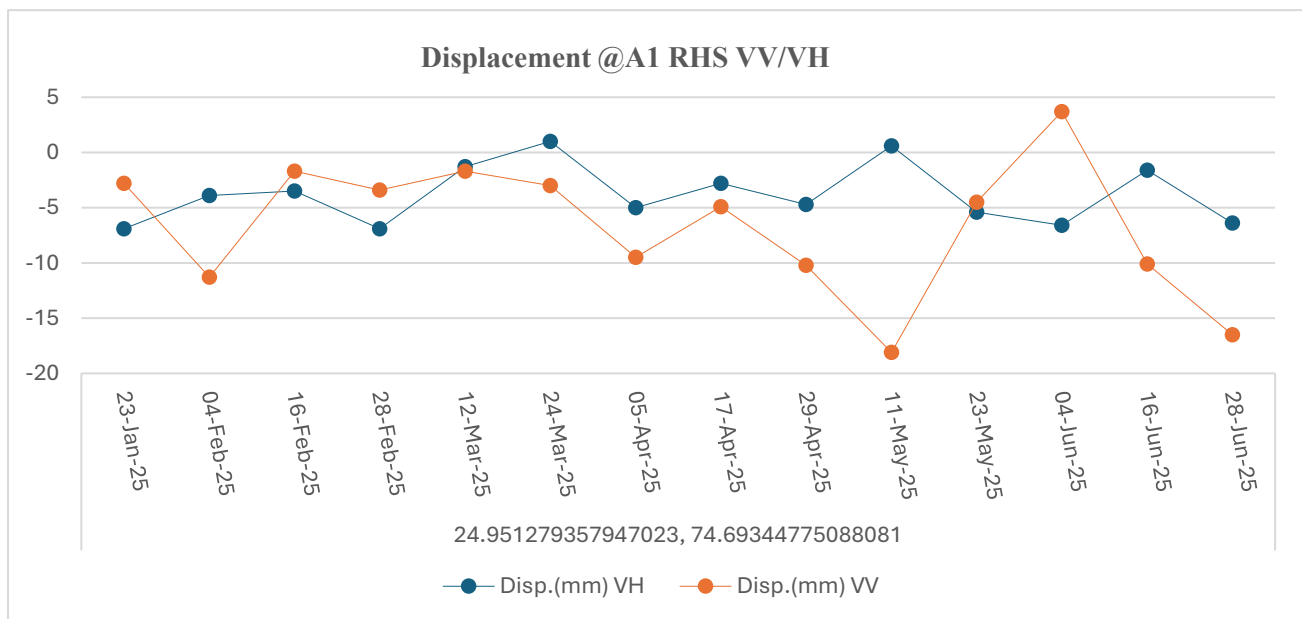
Critical Events & Anomalies:

- **January-February 2025:** VH transitions from slightly negative to a local positive peak around early February, while VV remains moderately positive, suggesting a temporary lateral adjustment under stable vertical conditions.
- **May 2025:** VH exhibits a secondary positive peak around mid-May, while VV decreases but remains positive, indicating short-term lateral flexibility with sustained uplift.
- **June 2025:** VH gradually increases again toward +7–8 mm, and VV stabilizes between +12 mm and +15 mm, suggesting post-peak stabilization with no progressive deformation.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (24.951279357947023, 74.69344775088081)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** remains relatively restrained, fluctuating approximately between -7 mm and $+1$ mm.
- **Vertical displacement (VV)** is predominantly negative, ranging roughly from -2 mm to -18 mm, indicating overall settlement-dominated behaviour with notable temporal variability.

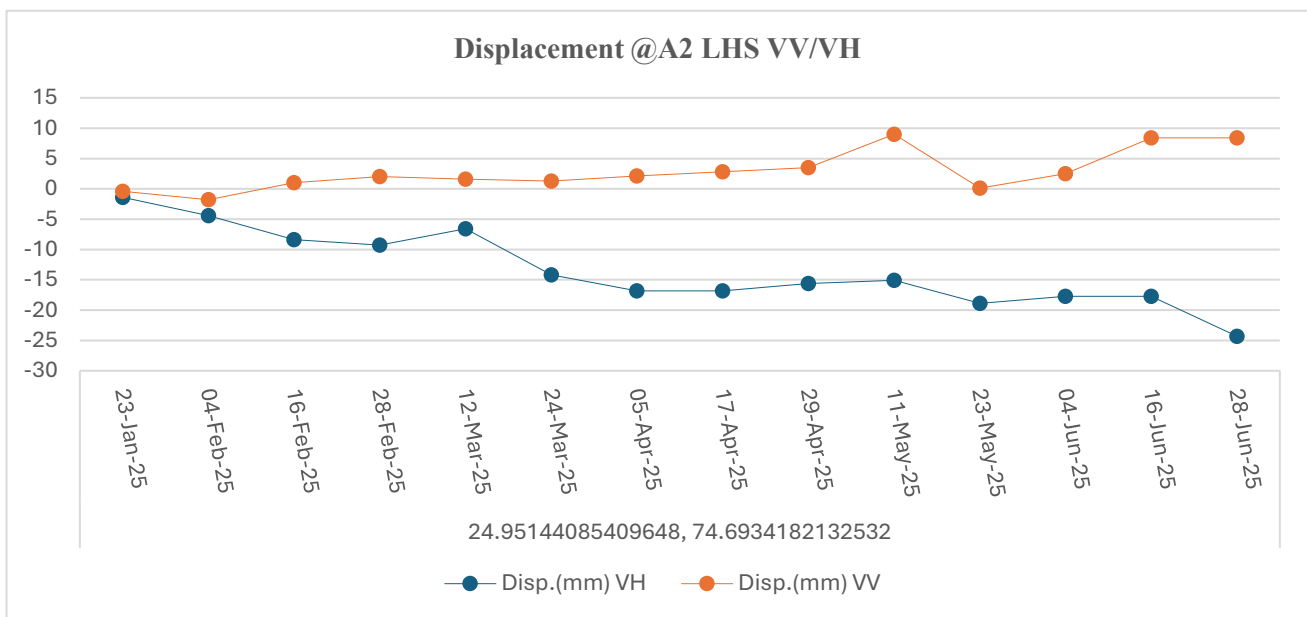
Critical Events & Anomalies:

- **January-February 2025:** VV shows an early downward excursion in early February, while VH remains mildly negative, suggesting initial settlement without lateral distress.
- **April 2025:** VV drops again to around -10 mm, with VH returning to approximately -5 mm, reflecting renewed settlement activity.
- **June 2025:** VV again trends downward to -16 mm, with VH stabilizing near -6 mm, indicating return to settlement-dominated behaviour.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (24.95144085409648, 74.6934182132532)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** exhibits strong negative dominance with progressive magnitude increase, ranging roughly between -2 mm and -27 mm.
- **Vertical Displacement (VV)** shows moderate variability with an overall positive bias, indicating dominant vertical uplift at the A2 LHS location throughout the monitoring period.

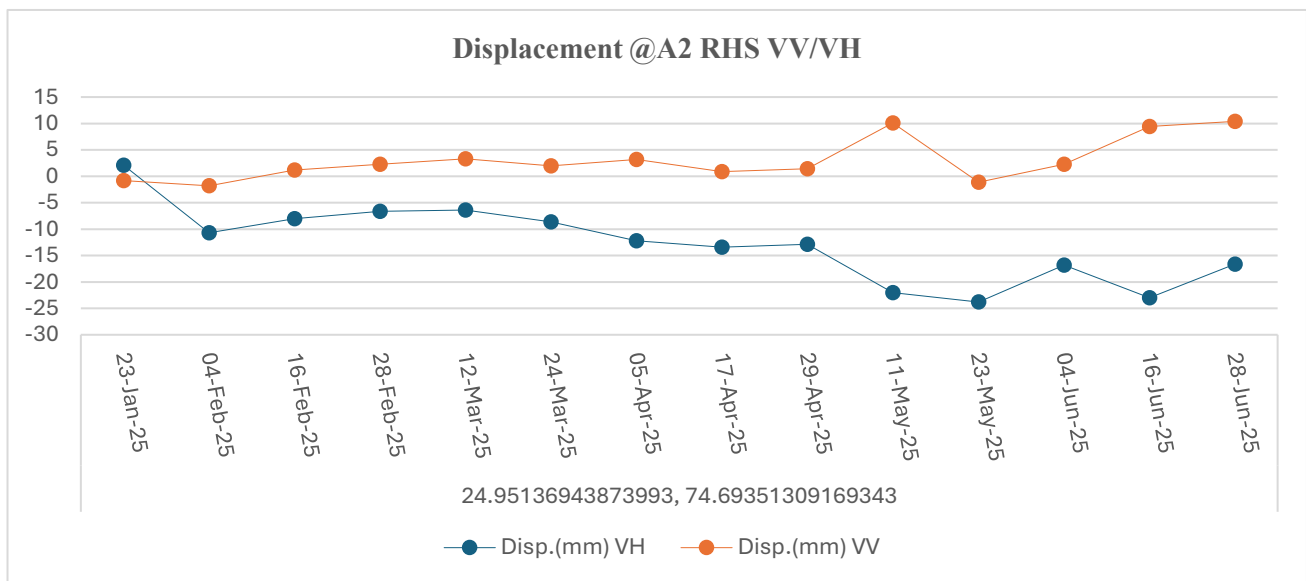
Critical Events & Anomalies:

- **January 2025:** VH begins near -2 mm, while VV is slightly positive ($+1$ mm), indicating initial mild lateral settlement with small vertical uplift.
- **June 2025:** VH reaches its most negative value (-27 mm), while VV increases to $+8$ mm, representing the most severe lateral displacement event of the monitoring period.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (24.95136943873993, 74.69351309169343)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** exhibits moderate variability with an overall positive dominance, indicating predominant vertical uplift at the A2 RHS location throughout the monitoring period.
- **Horizontal (VH) displacement** Horizontal displacement shows strong negative dominance with increasing magnitude, ranging roughly between +2 mm and -26 mm.

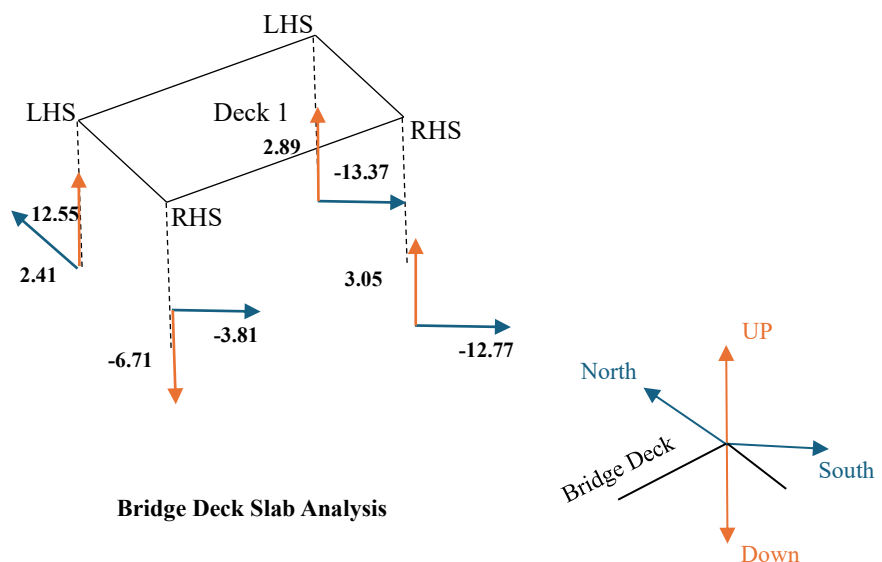
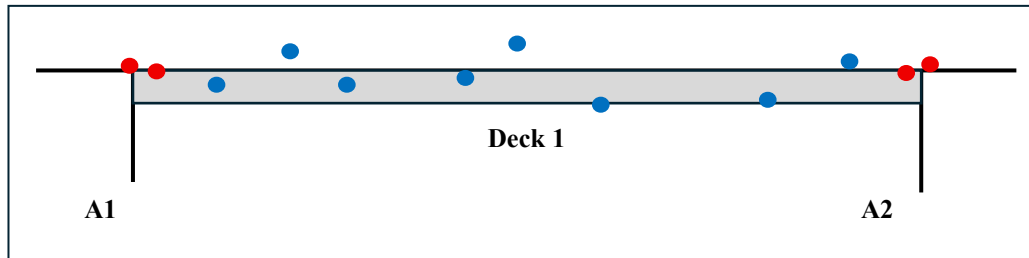
Critical Events & Anomalies:

- **January:** VH begins near +2 mm, while VV is slightly negative, indicating initial lateral neutrality with minor vertical settlement.
- **May 2025:** VH reaches its most negative value, while VV reduces to 0 mm, suggesting loss of uplift under intensified lateral displacement.
- **June 2025:** VH recovers slightly to -17 mm, while VV increases again to +2 mm, indicating partial post-event recovery.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- The bridge deck slab exhibits predominantly moderate vertical movements, with a mix of localized uplift and settlement across LHS and RHS regions. Overall, vertical displacement magnitudes remain relatively low, indicating stable vertical performance under prevailing conditions.

Horizontal Displacement Behaviour:

- Horizontal displacements are more pronounced than vertical movements, showing a clear directional bias across the deck. The RHS lower region records a horizontal displacement of approximately -12.77 mm, while the central lower region shows a smaller value of about -3.81 mm, indicating variable lateral restraint along the supports.

CONCLUSION:

- The bridge deck slab demonstrates a stable vertical response with limited uplift and settlement well within moderate limits. Horizontal displacements dominate the structural response, with significant asymmetry between LHS and RHS, highlighting potential torsional and differential lateral effects. Although no immediate structural concern is indicated, continued monitoring of lateral displacements, particularly at the deck edges and RHS lower region, is recommended to track any progressive trends.